

Derek P. Horkel

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Professional Summary

Hands-on Machine Learning Architect guiding algorithmic data infrastructure and regulated medical AI deployments. Expert in scalable ML operations, cross-functional delivery, and translating research innovations into resilient production systems built on ONNX Runtime and cloud-native platforms.

Education

University of Washington, Seattle, Washington USA

PhD in Physics, 2016

M.S. in Physics, 2012

University of Connecticut, Storrs, Connecticut USA

B.S. in Physics with Honors with minor in Mathematics, 2011

B.S. in Biological Sciences with minor in Ecology and Evolutionary Biology, 2011

Technical skills

Programming Languages: Python (statically typed), SQL, C/C++, Terraform, Bash, Node.js, R, Perl, SAS

ML & Data Science: PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, SciPy, PySpark, Jupyter, OpenCV, Peewee

MLOps & Platforms: ONNX Runtime, Pydantic, Structlog, ZenML, Kubernetes, Docker, CI/CD tooling, JFrog

Data & Cloud: AWS (S3, EMR, EC2, SageMaker, EKS, ECS, ECR, Batch, DynamoDB, RDS, Redshift, Lambda, CloudFormation, IAM, Rekognition), PostgreSQL, MySQL

Productivity: Mathematica, L^AT_EX, Microsoft Office, macOS, Linux, Windows

Languages: Spanish (elementary), Mandarin Chinese (elementary)

Employment

Machine Learning Architect, Digital Diagnostics

Jan 2026 – Present

- Led algorithmic data infrastructure spanning training data pipelines and compliance-ready datasets for regulated medical AI
- Delivered production preprocessing and ONNX Runtime inference framework, streamlining model deployment and integration, with structured logging and monitoring
- Worked with compliance and product stakeholders to create FDA-ready documentation and processes for regulated medical AI
- Built out integration between annotation platform, data repository, and active learning pipeline for optimizing data collection

Lead Machine Learning Software Engineer, Digital Diagnostics

Jan 2024 – Dec 2025

Senior Machine Learning Software Engineer

Nov 2022 – Dec 2023

- Built image and annotation governance services on Kubernetes and PostgreSQL, enabling traceable datasets across hundreds of thousands of patient submissions
- Introduced CI guardrails that decreased regression-related hotfixes across the ML stack
- Maintained FDA-cleared computer vision models for medical devices, coordinating field monitoring and retraining cadence with compliance teams
- Designed, deployed, and maintained a dataset registry backed by PostgreSQL and Kubernetes ingestion jobs to track submissions from dataset to exam
- Built reusable tooling for machine learning engineers, accelerating experimentation and deployment handoffs

Senior Machine Learning Engineer, Happy Health

Aug 2021 – Nov 2022

- Developed models using time series classification, regression, and Markov chain methods
- Ported Python ML models to C to run in iOS app
- Coordinated with firmware team to balance sensor data collection with battery demands

Principal Machine Learning Engineer, Sapient Industries

Jan 2021 – Aug 2021

Senior Software Engineer

Jun 2019 – Dec 2020

- Led end-to-end data collection, model development, and production deployment efforts across Sapient's energy analytics platform
- Iterated and retrained production models, improving accuracy while streamlining data ETL workflows
- Managed production, integration test, and development databases hosted in AWS RDS
- Brought AWS accounts into CIS benchmark compliance and implemented automated alerts
- Wrangled and cleansed data for client delivery and analytics use cases

Senior Machine Learning Engineer, Vanguard

Mar 2019 – May 2019

Machine Learning Engineer

Dec 2017 – Mar 2019

- Deployed, maintained and automated machine learning models and engineered features
- Handled data cleansing, wrangling, and staging for use in models
- Consulted on use of models in marketing campaigns, advising clients, and operations

Postdoctoral Fellow, Temple University

Oct 2016 – Oct 2017

Advisor Prof. Martha Constantinou

- Research focused on studying hadron structure using lattice quantum chromodynamics
- Worked with international collaboration coordinating and using computing resources

Research Assistant, University of Washington

June 2013 – Aug 2016

Advisor Prof. Stephen Sharpe

- Research focused on studying lattice quantum chromodynamics and effective field theories using statistical and numerical methods
- Used group theory along with numerical solvers to map out phase diagram of twisted mass lattice chiral perturbation theory
- Ran large scale Monte Carlo lattice simulation on the Hyak supercomputing cluster

Teaching Assistant, University of Washington

Sept 2011 – May 2016

- Designed and taught undergraduate section using Mathematica software for mathematical physics
- Taught introductory physics labs, tutorials, and exam grading
- Assistant for undergraduate and graduate quantum mechanics courses

Patents and Journal Publications

Powered device electrical data modeling and intelligence, US11681345B2 (2023)

Topological susceptibility from twisted mass fermions using spectral projectors and the gradient flow, *Phys. Rev. D* **97** (2018) 7, 074503

Phase structure with nonzero Θ_{QCD} and twisted mass fermions, *Phys. Rev. D* **92** (2015) 9, 094514

Impact of electromagnetism on phase structure for Wilson and twisted-mass fermions including isospin breaking, *Phys. Rev. D* **92** (2015) 7, 074501

Phase diagram of non-degenerate Wilson and twisted mass fermions, *PoS LATTICE2014* (2014) 066

Phase diagram of nondegenerate twisted mass fermions, *Phys. Rev. D* **90** (2014) 9, 094508